Cycle Test 2: worked solutions

15 Oct 2024 · 20 marks · 1 hour · CEPC14

Q1

3 marks

> 4 m^3 of M35 concrete is laid on a roof slab of the new civil annex building on 25-09-2023. Cube samples were made and tested after 7 and 28 days. Based on the results, comment on the strength and acceptability as per IS 456.

Specimen	7-day load (kN)	28-day load (kN)
CA 15/9/24 SP_1	505	721
CA 15/9/24 SP_2	685	982
CA 15/9/24 SP_3	635	907

Sampling: 4 m^3 falls in the 1–5 m^3 range → **1 sample** (3 cubes per test age), which matches what was cast.

Strengths (150 mm cubes, 22,500 mm^2):

Specimen	7-day (MPa)	Deviation	28-day (MPa)	Deviation
SP_1	22.44	**−17.0%**	32.04	**−17.1%**
SP_2	30.44	+12.6%	43.64	+12.9%
SP_3	28.22	+4.4%	40.31	+4.3%
Mean	27.04		**38.67**	

Comment:

- IS 456 Cl. 15.4: individual specimens must be within **±15% of the mean**. **SP_1 is 17% below the mean at both ages**, so this **sample (test result) is invalid** and cannot be used for acceptance.

- SP_1 is consistently low at 7 and 28 days, which points to a problem with that cube (poor compaction, damaged mould, wrong curing) or a weak batch.
- For information only: the mean of 38.67 MPa would pass the individual criterion ($\geq f_{ck} - 4 = 31$ MPa) but is below $f_{ck} + 4 = 39$ MPa, and SP_1 (32.0 MPa) is below f_{ck}.
- **Action:** since no valid result exists for this 4 m^3 pour, assess the in-place concrete by **rebound hammer / UPV and core tests** (IS 456 Cl. 17.4); the concrete is acceptable if the cores give an average equivalent cube strength of at least 85% of f_{ck} with no core below 75%.

Q2

5 marks

> Design an M50 concrete mix for bridge members. Use OPC 53 grade cement. Specific gravity of fine aggregate, coarse aggregate and cement are 2.65, 2.8 and 3.1. Water absorption of coarse and fine aggregate is 1.5%. Coarse aggregate of 20 mm rounded aggregate is only available. Placing by pumping (take slump according to pumping). Total moisture content in CA is 3% and in FA 1.5%. Specific gravity of superplasticizer is 1.35.

Assumptions: severe exposure (bridge in the open): max w/c 0.45, min cement 320 kg/m^3. Pumped concrete: **slump 100 mm** (IS 456 Cl. 7.1: 75–100 mm). Good supervision; zone II sand; SP 1% of cement with 20% water reduction; entrapped air 1%.

Step	Working	Result
Target strength	max(50 + 1.65 × 5, 50 + 6.5)	**58.25 MPa**
w/c	Fig. 1 of IS 10262:2019 (OPC 53) → 0.32 (< 0.45 ✓)	**0.32**
Water	(186 − 25 for rounded) = 161 → +6% for 100 mm = 170.7 → −20% (SP)	**136.5 kg/m^3**
Cement	136.5 / 0.32 (≥ 320, ≤ 450 ✓)	**426.6 kg/m^3**
SP	1% of cement	**4.27 kg/m^3**

Step	Working	Result
CA fraction	0.62 + (0.50 − 0.32)/0.05 × 0.01 = 0.656; × 0.9 (pumping)	**0.590**

Volume (m^3)	
Air	0.010
Cement 426.6 / 3100	0.1376
Water	0.1365
SP 4.27 / 1350	0.0032
All-in aggregate	**0.7127**
CA (SSD) = 0.7127 × 0.590 × 2800	**1178 kg**
FA (SSD) = 0.7127 × 0.410 × 2650	**774 kg**

Moisture correction: CA has 3 − 1.5 = **1.5% free water** → +17.7 kg water; weigh **1196 kg** wet CA. FA moisture equals its absorption (1.5%) → SSD, **774 kg**, no correction. Water to add = 136.5 − 17.7 = **118.9 kg**.

Final mix per m^3: OPC **427 kg**, water **119 kg**, FA **774 kg**, CA 20 mm **1196 kg**, SP **4.3 kg**; w/c 0.32; ratio **1 : 1.81 : 2.76** (SSD).

Q3

2 marks

> What is the need for mineral admixtures other than reduction of cement consumption? Generally mineral admixtures react with what in the case of ordinary Portland cement?

- Convert weak, leachable **$Ca(OH)_2$** into additional **C-S-H** → higher long-term strength.
- **Pore refinement** and denser ITZ → **lower permeability**, better resistance to chlorides, sulfates and ASR.
- **Lower heat of hydration** (mass concrete); better **workability** and less bleeding (fly ash).
- **Sustainability**: use of industrial by-products, less CO_2.

They react with the **calcium hydroxide** released by OPC hydration (pozzolanic reaction: CH + S + H → C-S-H).

Q4

2 marks

> Define workability of concrete. What are the tests to find workability? Why?

Workability: the ease of mixing, placing, compacting and finishing fresh concrete without segregation. Tests: **slump, compacting factor, Vebe, flow table, Kelly ball**. Several tests are needed because none measures workability directly and each suits a range: slump (medium–high, site), CF (low, lab), Vebe (very dry), flow table (flowing), Kelly ball (in-place check). Adequate workability ensures **full compaction** (5% voids cut strength by about 30%).

Q5

2 marks

> Select the size of aggregate for a concrete mix for a slab thickness of 50 mm with a spacing of reinforcement of 50 mm. What will be the limiting value of impact resistance for the aggregate?

Max size ≤ 1/4 × 50 = 12.5 mm (thickness) and ≤ 50 − 5 = 45 mm (bar spacing) → **10 mm aggregate**. Aggregate impact value **≤ 30%** if the slab is a wearing surface; **≤ 45%** otherwise (IS 383).

Q6

1 mark

> Limit for chlorides and sulphates content in water used for concrete mixing is ___ ppm and ___ ppm respectively.

Lecture values: **chlorides < 500 ppm**, **sulfates (SO_3) < 1000 ppm**. (IS 456: chlorides 500 mg/l for RCC, 2000 mg/l for PCC; SO_3 400 mg/l.)

Q7

1 mark

> If the prevailing temperature is above 40 °C and the concentration of sulphate in soil is more than 3, what type of cement will you use?

Sulphate resisting Portland cement (IS 12330). For sulfate classes above 3, IS 456 Table 4 allows only supersulphated cement or SRC, and **supersulphated cement must not be used where the temperature exceeds 40 °C**. Use SRC with min. cement 370 kg/m^3 and max. w/c 0.45 (Class 4), or 400 kg/m^3, w/c 0.40 plus a protective coating (Class 5).

Q8

1 mark

> What is the minimum cement content for very severe exposure condition and for 10 mm size aggregate?

380 kg/m^3 for RCC (340 from IS 456 Table 5 + 40 for 10 mm aggregate, Table 6); 300 kg/m^3 for plain concrete.

Q9

2 marks

> Design M15 concrete via nominal mix for fine aggregate of zone II and coarse aggregate of 20 mm size.

IS 456 Table 9: **350 kg dry aggregate per 50 kg cement**, FA : CA = 1 : 2, **32 l water per 50 kg cement** → per bag: cement 50 kg, FA **116.7 kg**, CA **233.3 kg**, water 32 l; ratio ≈ **1 : 2 : 4**; per m^3 ≈ cement 275 kg, FA 641 kg, CA 1282 kg, water 176 l.

Q10

1 mark

> Building needs to be built in a cold region (temperature below 1 °C). Which admixtures will you choose and why?

Non-chloride accelerator (calcium nitrate/formate) to gain strength before freezing; **air-entraining admixture** for freeze–thaw resistance; **antifreeze**

admixture to lower the freezing point. Avoid $CaCl_2$ in RCC (corrosion).